THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

Master of Business Administration (International)
IMBA6111 Data-Driven Strategic Decision Making

I. Information on Instructor

Instructor: Prof. Zhixi WAN (万智玺 教授)
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Zhixi Wan is a Professor in the area of Innovation and Information Management, in the Faculty of Business and Economics at the University of Hong Kong. Professor Wan teaches Business Analytics, Supply Chain and Logistics Management in the IMBA, the MsBA, and the DBA programs. Professor Wan holds a Ph.D. from the University of Michigan, and a Bachelor of Engineering from the Tsinghua University. His research focuses on supply chain management, value chain strategies, and platform economics and operations. Before joining HKU, in 2017 and 2018, he worked in Didichuxing, the leading ride-hailing platform firm, as a DiDi Economist and the head of the company's research center of Innovation and Operations Management.

Teaching assistants:

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II. Course Description and Objectives

Companies are creating and consuming vast amounts of information, leaving behind a trail of digitized big data. To be successful, companies want to extract value from data, and use them to support their decision-making, often under uncertainty due to incomplete information. However, managers often find it difficult to translate data into actionable policies, and generate business value. This course discusses both strategical-level and operational-level considerations. We start with an overview of (big) data, artificial intelligence (AI), their relationship, blessings and curses of AI, and how human beings can co-exist with AI for better future. Data alone cannot generate business value. We then discuss the analytics triangle of Problem – Data – Analytics for effective data-driven decision making, through case discussion and sharing from multiple industry sectors. We then introduce popular analytics techniques using real data, such as principal component analysis, clustering, logistic regression, time series forecasting, and so on. We use JMP - an user friendly, powerful analytical package. Assisted analysis of real data offer hands-on experience.

The course objectives are:

1. Obtain solid understanding about analytics in business situations
2. Formulate the right business problem and identify suitable analysis tools
3. Carry out decision making under uncertainty
4. Present analysis results in business relevant language
5. Understand ethical issues of data analytics

III. Course Learning Outcomes (CLOs)

CLO1	Clearly identify and formulate the relevant business problem
CLO2	Select and use effective methods to address the business problem
CLO3	Use software tools to provide solution to the issue at hand
CLO4	Communicate the solution effectively

IV. Alignment of Programme and Course ILOs

IMBA Programme Learning Outcomes	Aligned Course Learning Outcomes (CLOs)
PLO 1. Acquisition and internalization of business knowledge and skills in key functional areas	CLO1, CLO2, CLO3
PLO 2. Application of business knowledge to solve business problems	CLO1, CLO2, CLO3
PLO 3. Development of managerial leadership and inculcating professional ethics and competency in business	CLO1, CLO3, CLO4
PLO 4. Development of global outlook	CLO4
PLO 5. Mastering communication skills	CLO1, CLO2, CLO4

V. Teaching and Learning Activities (TLAs)

TLA#	Content	Aligned CLOs
TLA1	Lectures: Instructor will give lectures on major concepts and issues.	CLO1, CLO2, CLO3, CLO4
TLA2	Computer Workshops: Under the help of the instructor and teaching assistants, student will use software tools to solve mini business cases.	CLO1, CLO2, CLO3
TLA3	Case Competition and Presentation: Students will work in groups to solve a practical business case, and make managerial recommendations based on solid statistical analysis.	CLO1, CLO2, CLO3, CLO4
TLA4	Class participation: Students are encouraged to raise questions in class and actively participate in the class discussion.	CLO1, CLO2, CLO3, CLO4

VI. Assessment

A#	Content	Weight	Aligned CLOs
A1	Class Participation	20%	CLO1, CLO2, CLO3, CLO4
A2	Workshop Reports	30%	CLO1, CLO2, CLO3
A3	Case Competition	20%	CLO1, CLO2, CLO3, CLO4
A4	Individual Report	30%	CLO1, CLO2, CLO4

Class participation is an important component of the learning experience. It will be graded at a scale of 20. Active participation in the class is weighted more than class attendance. To maximize learning efficiency, quality of participation is more important than quantity. Please bear in mind the following criteria for comments/discussion:

- Relevance: Are the comments clearly related to the contents covered in class?
- Fact-based: Have you used specific facts to support your points?

- Logic: Is your reasoning logical and consistent?
- Originality: Do your comments bring fresh perspective to the class?

Groups will be asked to present selected articles or case study results to the class. Group presentation quality will be evaluated and graded as a component of class participation.

Workshop reports 1-3, each accounting for 10% of total course grade, are based on short questions related to computer workshops and the cases we will cover in the class. The questions (under Workshop Report sections) are available for download on Moodle. You should partner with at most two classmates and should submit one report stating all names on the cover page. Submissions should be made electronically on Moodle (the submitted file type should be .docx or .pdf. xlsx files will not be accepted). The submission deadlines are:

- Workshop report 1: midnight (23:59) on the day before Session 3
- Workshop report 2: midnight (23:59) on the day before Session 5
- Workshop report 3: midnight (23:59) on the day before Session 7

Late submissions will not be accepted.

The ***case competition*** will take place on the last day (Session 7 and 8). The deliverable is a presentation to be handed in later that day. Professor Wan and the teaching assistants will be available to assist with queries related to the theory or the software. Logistics details will be provided in Session 7.

The ***individual report*** is an individual assessment component that is due at midnight August 18, 2025. The report should propose a potential analytics project related to your work that can apply the machine learning methods and AI ethics analysis covered in the course. The proposal should be structured to answer the following questions:

1. Define the project, including the background, and how this analysis can be useful/beneficial.
2. Describe the data needed. Do you have the data ready? If yes, how can you retrieve the data (from whom in your organization)? If not, how do you plan to collect these data, and what are the potential difficulties?
3. Explain how you plan to conduct the analysis.
4. Explain how the results from the analysis can be used to draw conclusions and make business decisions.
5. Discuss the potential ethics issues in the implementation

The word limit is 1000 (not including graphs, tables, and reference). It should be submitted through Moodle before the deadline. If you have any difficulty submitting through Moodle, please send the report (in docx or pdf format) to Professor Wan via email (zhixiwan@hku.hk) before the deadline. The report will be graded based on relevance, clarity, depth and originality. All information sources used in the report should be clearly cited.

VII. Standard for Assessment

A+	Top 5% of the class based on weighted final numerical score
A, A-	Top 5% - 40% of the class based on weighted final numerical score
B+, B, B-	Students with a reasonably satisfactory understanding of the material covered in the class.
C	Bottom 5-10% of the class, but have achieved the minimum 50% in the weighted final numerical score
F	Students who fail to achieve the minimum 50% in the weighted final numerical score

VIII. Means/Processes for Student Feedback on Course

Feedbacks on workshop reports, group presentation, and case competition will be provided on Moodle or through email.

IX. Academic Integrity

The University Regulations on academic dishonesty will be strictly enforced. Please check the University Statement on plagiarism on http://www.hku.hk/plagiarism/.

Cheating or plagiarism of any kind would result in an automatic F grade for the course plus strict enforcement of all Faculty and/or University regulations regarding such behavior. Incident(s) of academic dishonesty will NOT be tolerated.

Academic dishonesty is a behavior in which a deliberately fraudulent misrepresentation is employed in an attempt to gain undeserved intellectual credit, either for oneself or for another. It includes, but is not necessarily limited to, the following types of cases:

a. Plagiarism - The representation of someone else's ideas as if they are one's own. Where the arguments, data, designs, etc., of someone else are being used in a paper, report, oral presentation, or similar academic project, this fact must be made explicitly clear by citing the appropriate references. The references must fully indicate the extent to which any parts of the project are not one's own work. Paraphrasing of someone else's ideas is still using someone else's ideas, and must be acknowledged.
b. Unauthorized Collaboration on Out-of-Class Projects - The representation of work as solely one's own when in fact it is the result of a joint effort.
c. Cheating on In-Class Exams - The covert gathering of information from other students, the use of unauthorized notes, unauthorized aids, etc.
d. Unauthorized Advance Access to an Exam - The representation of materials prepared at leisure, as a result of unauthorized advance access (however obtained), as if it were prepared under the rigors of the exam setting. This misrepresentation is dishonest in itself even if there are not compounding factors, such as unauthorized uses of books or notes.

GenAI policy: You are encouraged to use generative AI tools in this course, mirroring real business contexts where such tools are increasingly vital. For all assignments, please disclose the specific AI tools used and the scope of their contribution to your work. This transparency ensures academic integrity and prepares you for responsible AI integration in your professional career.

X. Course Material

The course is designed to be largely self-contained. All course materials, such as lecture slides, cases, articles, readings from popular press, computer workshops, data files, will be distributed digitally on HKU Moodle. A print course pack will be distributed by the program office.

Throughout the course, we will use the following book (Shmueli et al., 2023) as the technical reference:

- Shmueli, G., Bruce, P., Stephens, M., Anandamurthy, M. and Patel, N. (2023). ***Machine Learning for Business Analytics***. 2nd Edition. Wiley Global Research (STMS).

The book lays out key concepts about Machine Learning that are relevant to business analytics. It also provides many dataset examples with analysis in the software JMP Pro.

XI. Content and Tentative Teaching Schedule

Pre-class Preparation

- You need to be comfortable with simple mathematical notation and equations.
- Please bring your laptop with JMP SE installed
- Watch a 5-min short video to understand the relationship between AI and Machine Learning: https://youtu.be/4RixMPF4xis?si=ebd9NqzNRXYc-Cmy
- Read Colson (2019) "What AI-Driven Decision Making Looks Like," *Harvard Business Review* (General on Moodle)

Session 1: From Business Analytics to Strategic Decision Making

Format: Lecture

Concepts covered:

- Problem-Data-Analytics Triangle
- Strategic decision making in the era of AI
- Get familiar with JMP data visualization
- Use JMP to do regression analysis

Recommended readings:

- Shmueli et al., 2023: Chapter 3 (Data Visualization) and Chapter 6 (Multiple Linear Regression)

Before class:

- Download the data ***ToyotaCorolla1000.jmp*** (Session 1 on Moodle) to your computer. You can carry out the analysis on the file together with the instructor in class.

Session 2: Time-Series Analysis

Format: Lecture
Computer Workshop

Concepts covered:

- Components in a time series
- Forecasting with time-series models
- Smoothing methods
- Causal inference via time series methods

Recommended readings:

- Shmueli et al., 2023: Part VII: Forecasting Time Series (Chapters 17-19)

Before class:

- Download the data file ***Amtrak.jmp*** (Session 2 on Moodle) to your computer. You can carry out the analysis on the file together with the instructor in class.

After class:

- Complete Workshop Report 1
- Prepare "Wahoo Fitness: Segmentation and Data Insights", Darden case UV8688 (Session 3 on Moodle). Some questions in Workshop Report 1 are based on the case.

Session 3: Dimension Reduction and Cluster Analysis

Format: Lecture
Case Solving

Concepts covered:

- Principal Component Analysis (PCA)
- Clustering by k-means method

Recommended readings:

- Shmueli et al., 2023: Chapter 4 (Dimension Reduction)
- Shmueli et al., 2023: Chapter 16 (Cluster Analysis)

Before class:

- Be prepared to discuss Wahoo Fitness case

Session 4: Classification Problems (I)

Format: Lecture
Computer Workshop

Concepts covered:

- Predictive analysis by classification
- Evaluation of classifiers
- K-Nearest Neighbours (KNN)
- Logistic Regression

Recommended readings:

- Shmueli et al., 2023: Chapter 7 (k-Nearest Neighbours)
- Shmueli et al., 2023: Chapter 10 (Logistic Regression)

Before class:

- Download the data file ***UniversalBank.jmp*** (Session 4 on Moodle) to your computer. We will solve the case together in class.

After class

- Complete Workshop Report 2.
- Prepare "Agoda: People Analytics and Business Culture (A) and (B)", Ivey case W17429 (Session 6 on Moodle). Some questions in Workshop Report 2 are based on the case.

Session 5: Classification Problems (II)

Format: Lecture
Case Solving

Concepts covered:

- Classification trees
- Avoid overfitting
- Random forest and boosted trees

Recommended readings:

- Shmueli et al., 2023: Chapter 9 (Classification and Regression Trees)

Before class:

- Be prepared to discuss practice problems in Workshop Report 2

Session 6: AI Ethics

Format: Lecture
Case Solving
Computer Workshop

Concepts covered

- AI and talent management
- The big three ethical challenges in AI: bias; explainability; privacy
- Five ethical levels of privacy

Before class:

- Be prepared to discuss Agoda case

After class

- Complete Workshop report 3.

Session 7 & 8: Case Competition and Course Summary

Format: Lecture
Case competition between groups
Case presentation

Concepts covered

- Strategic decision making by classification methods
- User retention modelling
- Examples: user behavioural data analytics

Before class:

- Be prepared to discuss Workshop Report 3.
- Review the course material (especially those related to classification methods) to be prepared to solve the case in the case competition. Logistics of the case competition, and the case and data file, will be made available on the day.

(End)